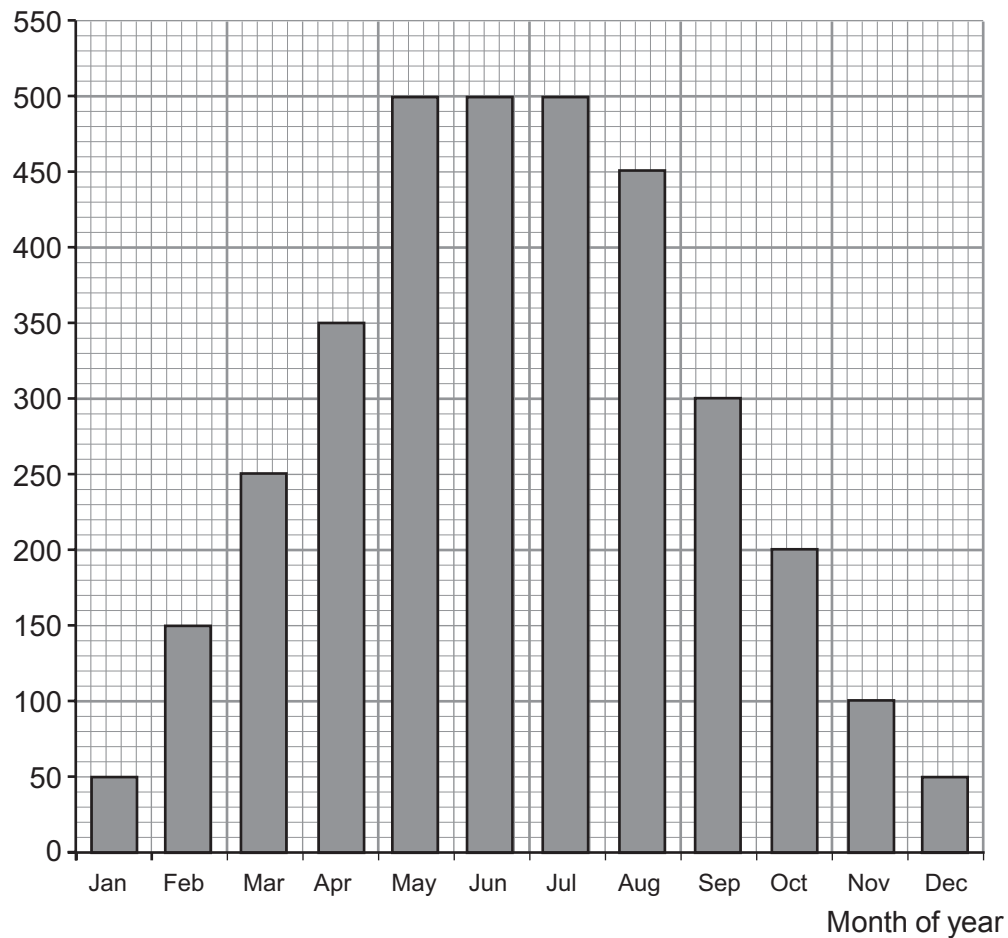


8. A family from Wales researched photovoltaic (PV) cells before they bought some. They were told by the manufacturer that they could expect the PV cells to generate 3400 kWh of electricity per year. **Figure 1** shows the energy generated in kWh from PV cells in a typical year in Wales. This was included in an advert for the PV cells.

Figure 1

Energy generated (kWh)



- (a) Use the information from **Figure 1** to validate the claim that PV cells could generate 3400 kWh annually. [2]

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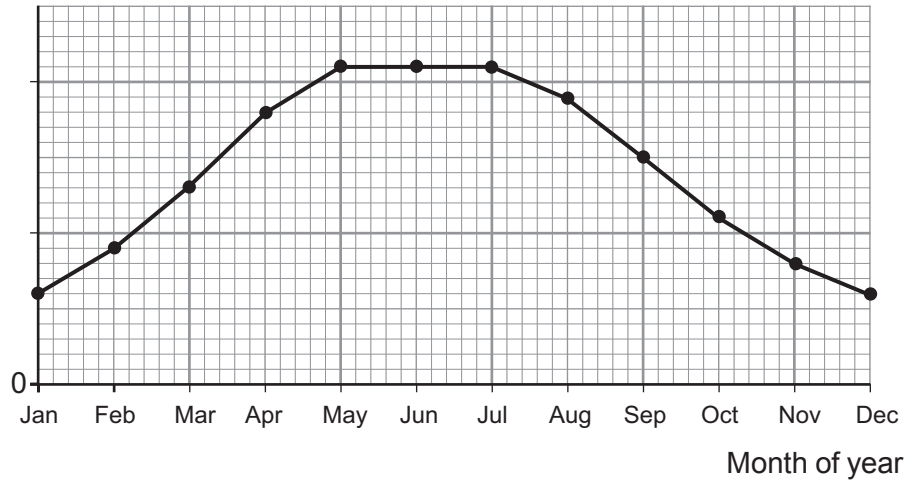
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- (b) Sunshine data for Wales over the last 30 years was obtained from the Welsh MET weather office. **Figure 2** shows the mean number of sunshine hours for each month in Wales.

Figure 2

Mean number of
sunshine hours



- (i) Describe how the information from **Figure 2** supports the general trend shown in **Figure 1**. [1]

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- (ii) The Welsh MET weather office also had data for the **maximum** number of sunshine hours for each month. State why the manufacturers were **not** allowed to use this data in their advert. [1]

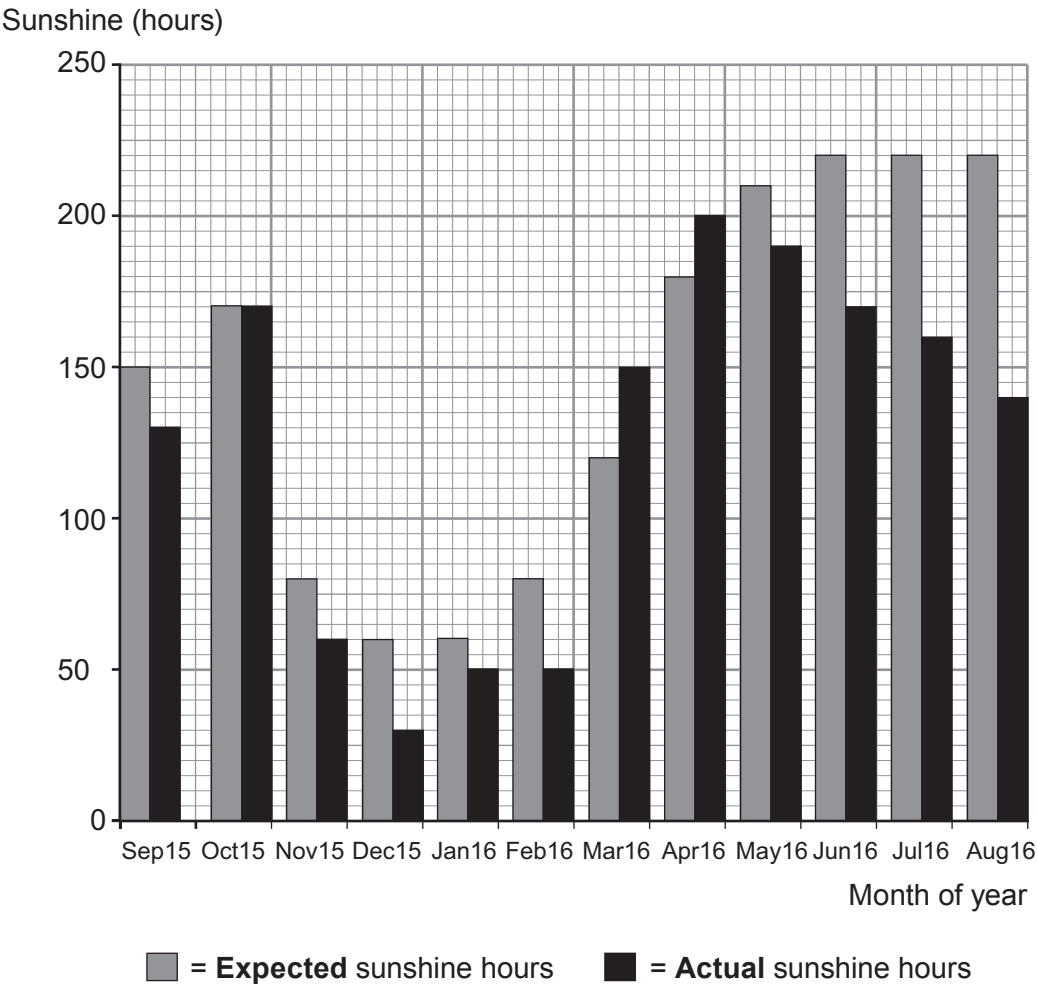
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- (c) In September 2015 the family had PV cells installed on their south facing roof. They collected 12 months of actual data from their PV cells. **Figure 3** compares the **expected** sunshine hours with **actual** sunshine hours. The **expected** sunshine hours data were obtained from local weather station records.

Figure 3



- (i) Use information from **Figure 3** to tick (✓) the **three** correct statements. [3]

In Dec15 the **actual** number of sunshine hours was double the **expected** number.

☐

Apr 16 was the month that had the most **actual** sunshine hours.

☐

June, July and August have different **expected** sunshine hours.

☐

Jan 16 had the least number of **actual** sunshine hours.

☐

Apr 16 had 4 times the **actual** number of sunshine hours compared to Jan 16.

☐

There is only one month where the **actual** and the **expected** sunshine hours were the same.

☐

- (ii) In April 16 the PV cells generated 600 kWh. The manufacturers claim the PV cells will produce 3 kW in sunshine. Use an equation from page 2 and **Figure 3** to validate this claim. [2]

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- (d) The PV cells were expected to produce 3400 kWh of electricity. However, the total number produced was 3670 kWh.

- (i) The family saves 29 p for each kWh of energy generated. Calculate how much extra money they saved. [2]

Extra savings = p

- (ii) If the PV cells continue to generate more energy than expected what effect would this have on their payback time? [1]

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END OF PAPER